

Tensile properties and pullout behaviour of AFRP and CFRP rods for grouted anchor applications

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Abstract

Fiber-reinforced-plastic (FRP) rods have been introduced in the market for grouted ground anchors. The resistance to corrosion and chemical attack, high strength-to-weight ratio, and ease of handling of these rods make them a better alternative to steel tendons in some applications of ground anchors. However, to fully utilise FRP rods as tendons for cement grouted anchors, some aspects of their behaviour have to be determined, including tensile properties, tensile capacity, bond strength in cement grout, long-term strength and durability in alkaline environments. In this paper, the tensile characteristics, bond strength and pullout behaviour of AFRP and CFRP rods embedded in cement grout are discussed. The pullout test program involved four types of rods, four types of cement grouts and three types of anchorage tubes with three bonded lengths. The experimental results have shown that the tensile properties of the FRP rods are close to the manufacturers' data. The surface geometry of FRP rods, the properties of the filling grout and the stiffness of the host medium (anchorage tube) influence the pullout behaviour, pullout capacity and maximum bond stress of cement grouted FRP anchors. © 2000 Elsevier Science Ltd. All rights reserved.

Keywords: Fiber-reinforced-plastic (FRP); Anchor; Pullout capacity

1. Introduction

Fiber-reinforced-plastic (FRP) rods are currently being used as reinforcing bars, prestressing tendons, and tensile elements for post-tensioned ground anchors, mostly in Japan, Europe and in North America [1–6]. FRP rods are not prone to corrosion which is one of the major disadvantages of steel. They can also have advantages over steel in being of lighter weight and having higher tensile capacity, and electro-magnetic neutrality.

Unlike steel, the pultrusion process by which FRP rods are manufactured results in rods with a fairly

constant cross-section and a smooth surface. Surface indentations on the rod are produced by wrapping a fiber string around the rod before the resin sets. Moreover, some manufacturers add a coating of sand and resin on the rod at the end of the manufacturing process to make the rod's surface rough. Furthermore, the stress–strain relationship of FRP rods is fairly linear at all stress levels up to the point of failure, without exhibiting any yielding characteristics like that observed for steel. As the surface deformation and mechanical properties of FRP rods are different from those of steel tendons, sufficient experimental information about the tensile performance, bond properties and pullout capacity of FRP rods embedded in grout are required to develop design guidelines.

The objectives of the project reported in this paper were: (1) to evaluate the tensile properties (strength, stiffness and maximum elongation) of some commonly

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used FRP rods; and (2) to investigate the pullout behaviour and bond strength of those rods embedded in cement grout. Pullout capacity, maximum bond stress and load-slip relationship were obtained from laboratory tests. The experimental program included four types of aramid FRP (AFRP) and carbon FRP (CFRP) rods with an identical nominal diameter of 8 mm, four types of cement-based grouts and three types of anchorage tubes (host media).

1.1. Description and properties of AFRP and CFRP rods used in the study

FRP tendons are available in the form of rod or cable, rectangular strip, braided rod, and multi-wire strand. They are commonly used in prestressed concrete structures and post-tensioned ground anchors. Currently, only AFRP (aramid) and CFRP (carbon) tendons are being used for FRP ground anchors [6]. In this project, four types of commercially available FRP rods, namely Arapree, Technora, Leadline and CFCC, were selected for the investigation. Their general characteristics and compositions (fibers and resins) are briefly introduced.

Arapree (ARAamid PREstressing Element) was developed jointly by AKZO and Hollandische Beton Groep in the Netherlands, and manufacturing rights have been transferred to Sireg SPA of Italy [7]. Arapree is made of Twaron HM aramid fibers impregnated with epoxy resin using a pultrusion process. The composition is 45% Twaron HM fibers and 55% epoxy resin by volume. Arapree tendons are available in rectangular strips with cross-sections of 1.5×20 , 3.0×20 and 6.0×20 mm, and in circular rods with diameters of 5, 7.5 and 10 mm. The surface of the rods is sand coated to improve bond. Recently, a new Arapree based on Twaron IM fibers was introduced [8]. The 7.5 mm and 10 mm diameter round rods with sand-coated surfaces were used in the study (Fig. 1).

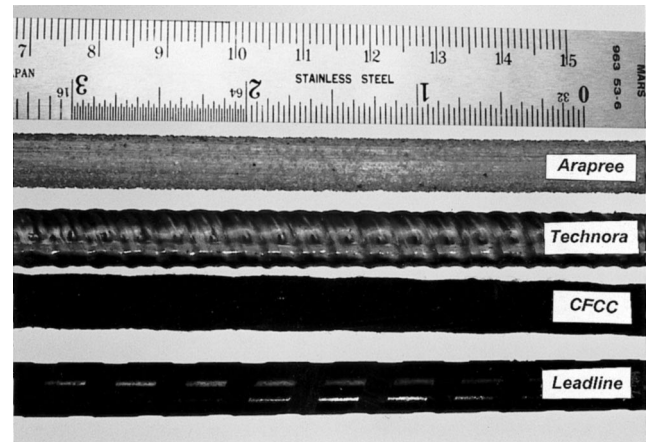


Fig. 1. Overview of FRP rods used in the study.

Technora was developed jointly by Sumitomo Construction Company and Teijin Corporation in Japan [9]. Technora is fabricated by a pultrusion process and is composed of Technora aramid fibers impregnated with vinyl ester resin. The contents of fiber and resin are 65 and 35% by volume, respectively. Both plain and wound rods with externally spiral windings are available. The available diameters of Technora rods are 3, 4, 6, 7.4 and 8 mm. Twisted seven-wire strands of 12.4 mm diameter are also available. An 8 mm diameter spirally-wound rod was selected for the study (Fig. 1).

Leadline is manufactured using a pultrusion method by the Mitsubishi Chemical Corporation in Japan [10]. Leadline is composed of pitch-based carbon fibers and epoxy resin. The contents of fiber and resin are 65 and 35% by volume, respectively. Leadline rods are available in round, indented, or rib shapes. The round rods have no surface deformation. The indented rods have two shallow helical depressions on the surface that spiral in opposite directions. The ribbed rods have either raised helical windings similar to the indented pattern or a circumferential winding transverse to the

Table 1
Manufacturer's data for FRP rods^a

Tendon	Type	Nominal diameter (mm)	Cross-section (mm ²)	Mean tensile strength (MPa)	Elastic modulus (GPa)	Ultimate strain (%)	Density (g/cm ³)	Poisson's ratio
Arapree-8	f200000	7.5	44.2	1506 (1370)	62.5	2.40	1.25	0.38
Arapree-10	f350000	10	78.5	1365 (1220)	62.5	2.40	1.25	0.38
Technora	Spiral wound	8	50.2	2140 (1900)	54	3.70	1.30	0.35
Leadline	Indented	7.9	46.1	2550 (2250)	150	1.30	1.67	n.a.
CFCC	1 × 7 × 7.5	7.5	30.4	2120 (1870)	137.3	1.57	2.10	n.a.

^aNotes: n.a., non-available; and values in parentheses are characteristic strength (the minimum guaranteed strength).

longitudinal axis of the rod. The available diameters of the rods are 3, 5, 8, 10, 12 and 17 mm. In this project, an indented 8 mm diameter rod was used (Fig. 1).

CFCC (Carbon Fiber Composite Cable) was developed by Tokyo Rope in Japan [11]. CFCC is composed of PAN type carbon fibers impregnated with epoxy resin. The cable is formed by twisting a number of small diameter rods similarly to a conventional strand. The contents of fiber and resin are 64 and 36% by volume, respectively. CFCC is available in single rods of 1.5 and 5 mm in diameter, or 7, 19, and 37 wire strands with a diameter ranging from 5.0 to 40.0 mm. The seven-wire strand with a nominal diameter of 7.5 mm was used in this investigation (Fig. 1).

FRP rods are anisotropic materials and factors such as type and volume of fiber and resin, fiber orientation and quality control during the manufacturing play a major role in the mechanical characteristics of the product. The matrix with its low strength does not significantly contribute to the strength of the rod. Hence the strength and modulus of elasticity of an FRP depend mainly on the properties of the fiber and its volume ratio. Table 1 shows the physical and mechanical properties of the FRP rods from manufacturers. It can be seen that the characteristic tensile strength and modulus of elasticity of Arapree, Technora, Leadline and CFCC are 1220 MPa and 62.5 GPa, 1900 MPa and 54 GPa, 2250 MPa and 150 GPa, and 1870 MPa and 137.3 GPa, respectively. The tensile strength of FRP rods is 3.0–5.6 times that of conventional steel rods (the yield strength is 400 MPa), while the modulus of elasticity of FRP rods is 28–77% that of steel rods (the modulus of elasticity is 195 GPa).

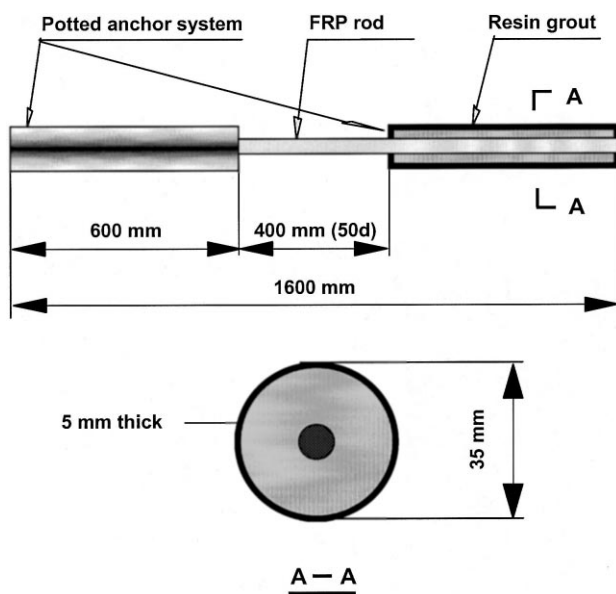


Fig. 2. FRP tensile specimens.

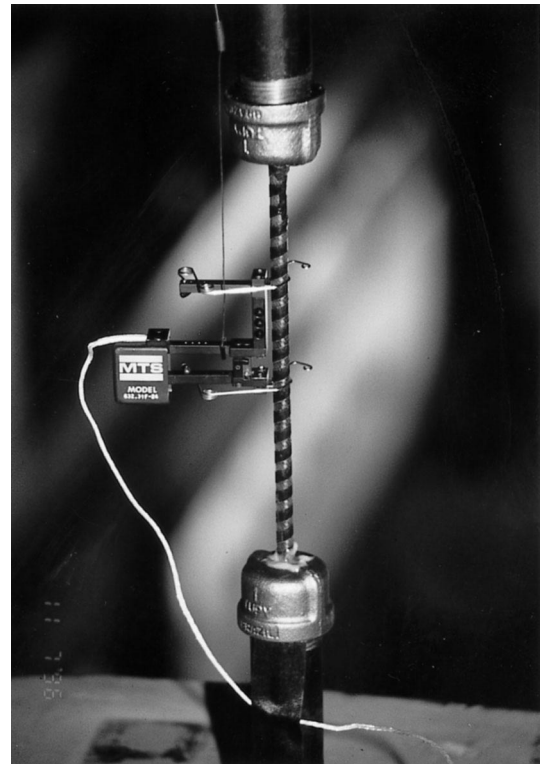


Fig. 3. Overview of the FRP rod in tensile tests.

2. Experimental program

2.1. Tensile tests

The aforementioned four types of FRP rods were selected for tensile tests. Each specimen was cut into 1600-mm long sections and anchored with a potted anchor system at each end. This method bonds the rod to the steel cylinder utilizing a mortar (resin or cement-based grout). The potted anchors used in this project consisted of 600-mm long steel cylinders filled with a high performance resin grout as shown in Fig. 2. Tensile tests were carried out using a hydraulic loading machine, 'Universal Testing Machine', having a capacity of 270 kN. The specimen was inserted into the test frame and gripped by the two thick plates at the anchored ends. The internal load cell was used to monitor the applied loads. Two clip-on 50-mm extensometers were attached to the specimen at the onset of the test and used to measure the longitudinal and transverse deformations of the specimen by means of built-in attachment springs (Fig. 3). The load cell and the extensometers were connected to a data logger system. The tension force was applied at a constant specification loading rate of 250 MPa/min [12].

2.2. Pullout tests

The objective of pullout tests was to investigate the pullout performance and the effects of different

Table 2
Summary of physical and mechanical properties of cement grouts^a

Grout type	Cement type/ components	Temp (°C)	Water bleed. (%)	Expan. (%)	Cone flow time (s)	Cohesion (MPa)	Friction angle (°)	Compressive strength (MPa)	Tensile strength (MPa)	Elastic modulus (GPa)	Poisson's ratio (ν)
CG1	Type 10	21	1	–	51.9	16.1 [7]	31.0 [7]	62.6 ± 0.1 (4)	3.3 ± 4.0 (5)	17.4 ± 0.9 (6)	0.11 ± 0.00 (6)
CG2	Type 30	23	0.74	–	58.1	16.9 [7]	31.8 [7]	63.5 ± 2.6 (4)	3.8 ± 0.3 (5)	18.6 ± 2.0 (6)	0.11 ± 0.01 (6)
CG3	CG1 + sand	22	1	–	64.8	14.0 [7]	33.8 [7]	57.7 ± 2.8 (4)	4.1 ± 0.2 (5)	22.9 ± 1.4 (6)	0.10 ± 0.01 (6)
CG4	Type 10 SF + swelling agent	21	–	5	53.2	12.0 [7]	36.5 [7]	53.5 ± 3.8 (4)	4.5 ± 0.5 (5)	16.7 ± 0.0 (6)	0.12 ± 0.00 (6)

^aNote: () number of tests, [] number of specimens.

Table 3
Geometric and mechanical characteristics of anchorage tubes^a

Anchorage tube made of	r_2 (mm)	r_3 (mm)	E_m (GPa)	ν_m	E_r (GPa)	ν_r
Steel	25.5	28.5	195	0.3	28.33	0.25
Aluminium	22.3	25.3	79	0.3	12.93	0.25
PVC	20.3	24.3	1	0.3	0.23	0.25

^aNotes: r_2 , r_3 , E_m , and ν_m = the inner diameter, outer diameter, modulus of elasticity and Poisson's ratio of the tube, respectively; E_r , ν_r = the modulus of elasticity and Poisson's ratio of the equivalent rock medium, respectively.

parameters on the bond characteristics of the FRP rod to cement grout. The variables for the tests were rod type, grout type, bonded length and anchorage tube type (host medium). Four types of FRP rods were used for the pullout tests. The FRP rods were those selected for the tensile performance tests.

Four types of cement-based grouts CG1 to CG4 were used as bonding agents with and without additives including swelling agent, superplasticizer, silica fume and sand. The water–cement ratio was 0.4 for all the cement grouts used in this study. Cement grout CG1 was a plain grout made from Type 10 Portland cement (ASTM I) with 0.6% of superplasticizer solids by mass of cement. Cement grout CG2 was also a plain grout but was made from Type 30 Portland cement (ASTM II) with 1% of superplasticizer solids by mass of cement. Cement grout CG3 was made by adding 40% of sand by mass of cement to CG1. Finally, grout CG4 was made from Type 10 SF cement (Blended Type I cement containing 8% silica fume) and included 1% of anti-bleed agent, 0.004% of swelling agent and 0.6% of superplasticiser solids by mass of cement. The physical and mechanical properties of the four types of grout are summarised in Table 2.

Three types of cylindrical tubes, steel, aluminium and poly(vinyl chloride) (PVC), were used as host media to investigate the effect of the host medium on the bond behaviour of grouted FRP anchors. The geometric and mechanical characteristics of the tubes are shown in Table 3. From the theory of thick wall cylinders, the modulus of elasticity of the equivalent rock medium with a Poisson's ratio of 0.25 is 37.6, 16.9 and 0.3 GPa for steel, aluminium and PVC tubes, respectively. The single FRP rod was inserted into the tube and bonded with a cement grout. Three bonded lengths of 100, 200 and 350 mm were selected for the anchor models. The bond-free length of 50 mm above the bonded length was obtained by covering the rod with a thin plastic tube as shown in Fig. 4. Four specimens of each anchor model were tested at 28 days after grouting.

Pullout tests on the anchor models were carried out using the hydraulic loading machine as used for tensile tests. The rate of loading was fixed at 22 kN/min. The pullout load was measured using an internal load cell.

The displacement of the rod was measured using two Linear Voltage Differential Transfomers (LVDTs) fixed at the loaded and free ends, respectively. The load cell and LVDTs were connected to a data acquisition system. Fig. 5 shows the set-up of laboratory pullout tests.

3. Test results and discussion

3.1. Tensile properties of FRP rods

All tensile tests of the FRP rod specimens were conducted until rupture of the FRP rods. Fig. 6 shows the typical tensile failure phenomenon of FRP rods Arapree, Leadline and CFCC. The failure occurs outside of the potted anchor systems. As shown in Fig. 7, typical stress–strain curves of FRP rods are practically linear at all stress levels up to the point of failure, without exhibiting any yielding like that observed for steel. The slope of the curve corresponded to the experimental modulus of elasticity (E). The abscissa of the maximum point was then used to estimate ultimate strain at failure. The tensile test results for all specimens are summarised in Table 4. The comparison results of experimental vs. manufacturers' data [9,13–15] indicate that:

3.1.1. AFRP rod Arapree

The three tested specimens obtain an average ultimate load of 63.03 kN, 5% lower than that of the reported Arapree rods. For the first specimen, the

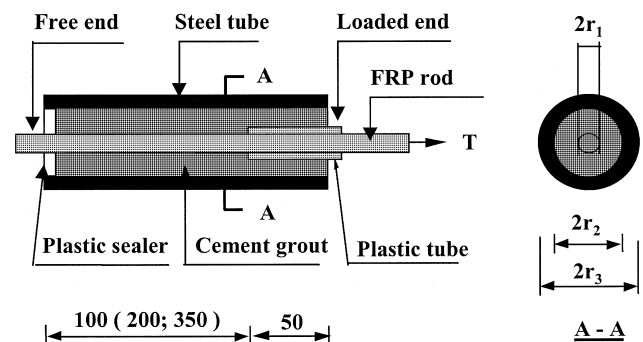


Fig. 4. Grouted FRP anchor models.

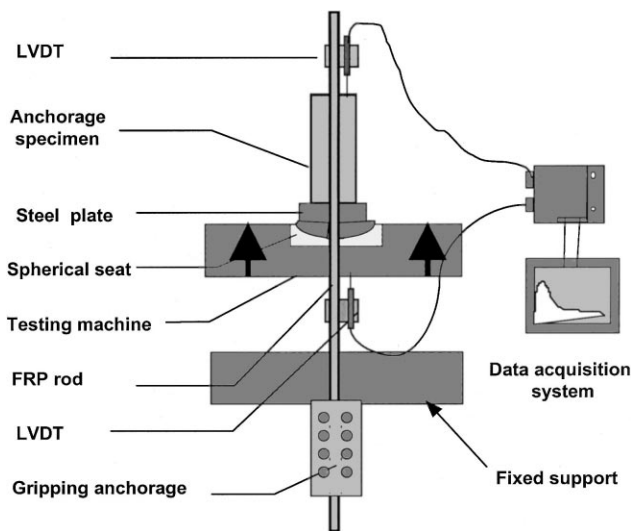


Fig. 5. Set-up of laboratory pullout tests.

recorded breaking load is 4% lower than the reported mean and the extrapolated ultimate strain is 2% higher than the reported value. The second specimen records a failure load 8% lower and an extrapolated ultimate strain 6% lower, than the reported values, respectively. The third specimen behaves much the same as the first one, recording a failure load 4% lower than the reported mean. The estimated maximum strain is 0.0234, which is lower than the reported ultimate strain by 2.5%. The values of the modulus of elasticity are 5, 1.6 and 1.1% below the reported value for specimens 1, 2 and 3, respectively.

3.1.2. AFRP rod Technora

The results of breaking load, stress and strain of Technora rod specimens are not reported in Table 4

Table 4

Summary of experimental tensile test results^a

Tendon type	Data source	Ultimate load (kN)	Ultimate strain (%)	Young's modulus (GPa)	Poisson's ratio
Arapree-8	Reported [13,14]	66.6 (60.8)	2.40	62.5	0.38
	Test 1	64.1	2.45	59.2	0.38
	Test 2	61.1	2.25	61.5	0.37
	Test 3	63.9	2.34	61.8	0.38
Technora	Reported [9]	(86.2)	3.7	54.0	0.35
	Test 1	—	—	44.3	—
	Test 2	—	—	42.7	—
Leadline	Reported [13]	> 120 (104)	1.30	150.0	n.a.
	Test 1	135.0	1.80	168.0	—
	Test 2	134.0	1.70	162.0	—
	Test 3	133.0	1.86	160.0	—
CFCC	Reported [15]	63.4 (56.9)	1.57	137.3	n.a.
	Test 1	69.2	1.60	140.6	—
	Test 2	65.7	1.56	138.7	—

^aNotes: n.a. = non-available; and values in parentheses are the guaranteed capacity.

because the rod could not fail in tension due to the continuous slip of the rod to the grout. The rod/anchor system used is only capable of attaining approximately 40% of the reported ultimate load. However, the moduli of elasticity of the two specimens are determined. The obtained values are 18 and 21% lower than the reported modulus, respectively.

3.1.3. CFRP rod Leadline

For the three tested specimens, tensile failure occurs at a higher load level than the reported failure limit, 10% more than the reported mean and 30% more than the guaranteed capacity. The obtained average maximum strain and modulus of elasticity are 37 and 9% higher than the reported values, respectively.

3.1.4. CFRP rod CFCC

The experimental data for the tested CFCC specimens 1 and 2 are higher than those reported by the manufacturer by 21 and 15% in guaranteed capacity, and 2.4 and 1.0% in modulus of elasticity, respectively. The mean ultimate load of the two tested specimens is 5% higher than the reported value. The maximum strains of the two specimens show almost the same level as the reported value.

3.2. Bond behaviour of cement grouted FRP anchors

3.2.1. Expression of test results

The pullout behaviour of cement grouted anchors was expressed by load–slip curves, characterized by the maximum and residual bond stress, and initial stiffness of the anchors as shown in Fig. 8 and defined as follows:

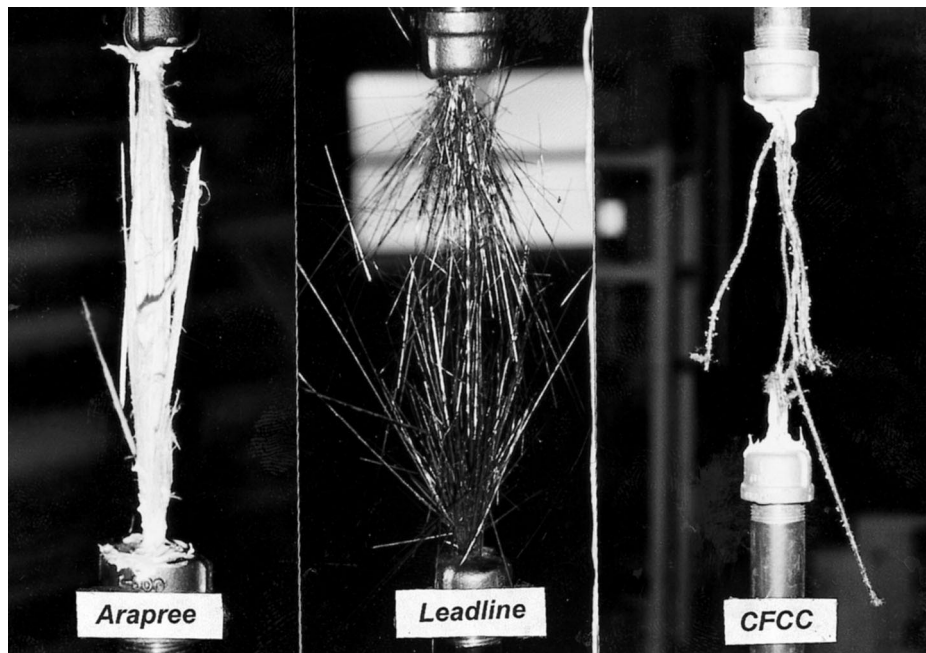


Fig. 6. Typical tensile failure of FRP tendons.

Slip. The slip along the bonded length is expressed with the net displacement after taking into account the elastic elongation of the rod:

$$\delta = \delta_{\text{mea}} - \delta_{\text{rod}} \quad (1)$$

where δ = rod slip along the bonded length; δ_{mea} = measured total displacement; and δ_{rod} = rod elastic elongation.

The values of δ_{mea} were measured continuously with

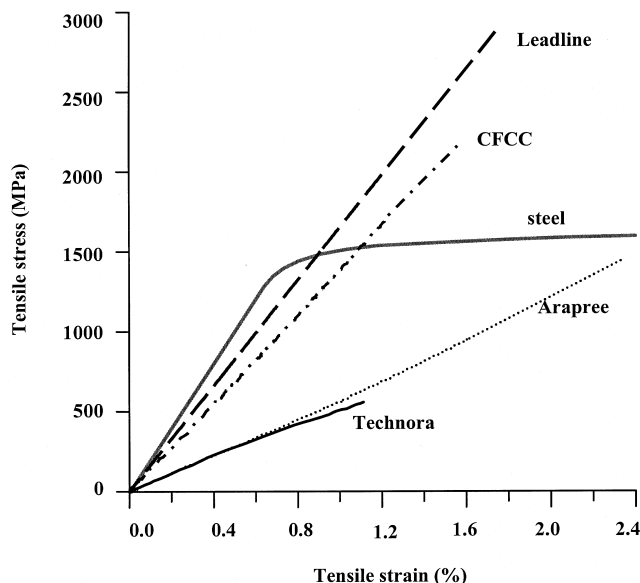


Fig. 7. Typical tensile stress-strain behaviour of FRP and steel tendons.

the corresponding loads by the LVDTs. The rod elongation is calculated from $\delta_{\text{rod}} = (TS)/(EA)$, where T is the applied load; S is the distance between the top of the bonded length and the measuring point of the LVDT; E is the modulus of elasticity; and A is the cross-sectional area.

Bond stress (τ). Bond stress (shear stress) refers to the shear resistance per unit contact surface and is obtained by dividing the pullout resistance force by the surface area of the rod in contact with the grout with the assumption of uniform bond stress distribution along the bonded length:

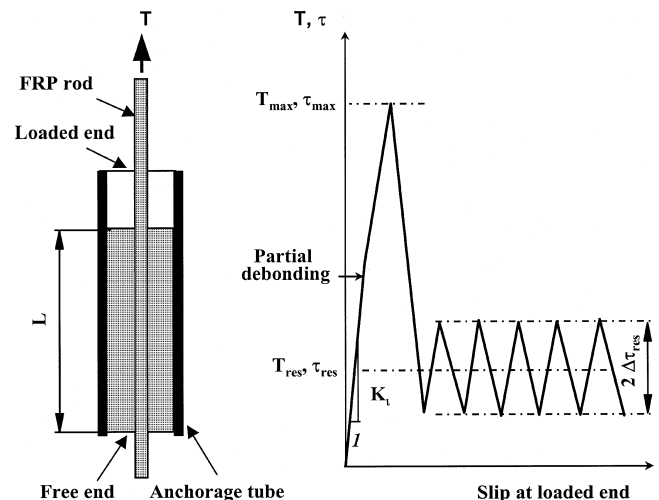


Fig. 8. Determination of bond properties of grouted FRP anchors.

$$\tau = \frac{T}{\pi dL} \quad (2)$$

where T = applied tensioning load; d = rod diameter; and L = bonded length of the rod.

Initial stiffness (k_i). Initial stiffness is the slope of the load–slip curve at the loaded end from the original point and is expressed by:

$$k_i = \frac{T}{\delta} \quad \text{or} \quad k_i = \frac{\tau}{\delta} \quad (3)$$

3.3. Pullout load–slip behaviour

Typical relations between the applied load and corresponding slip of the FRP rods Arapree and Leadline grouted in steel tubes are shown in Fig. 9a,b. The load–slip curves generally comprise ascending (pre-peak) and descending (post-peak) phases. In the ascending phase, the pullout load increases significantly for a given increase in slip at the loaded end until the peak load is reached. The load transfer from FRP rod to grout depends mainly on the adhesion and mechanical interlocking. The latter contributes most to the total load for rods with a deformed surface. The ultimate pullout capacity occurs once the slip begins at the free end, while the corresponding slip at the loaded end has apparently happened. This indicates that the bond stress distribution at the rod–grout interface is not uniform along the bonded length. The peak bond stress initiating at the loaded end of grouted anchors introduces a progressive bond collapse propagating toward the tail of the anchor.

The bond collapse of grouted FRP anchors is due to the detachment of the fiber spiral or sand grains, or due to the longitudinal shear of the surface deformation of the rod or the injected grout. FRP failure typically governs the FRP/grout bond. In the descending phase, grouted FRP anchors exhibit a sharp decrease in pullout capacity with an increase in slip and thereafter present a lower and oscillating residual pullout capacity. This implies that the residual pullout capacity depends not only on the friction resistance but

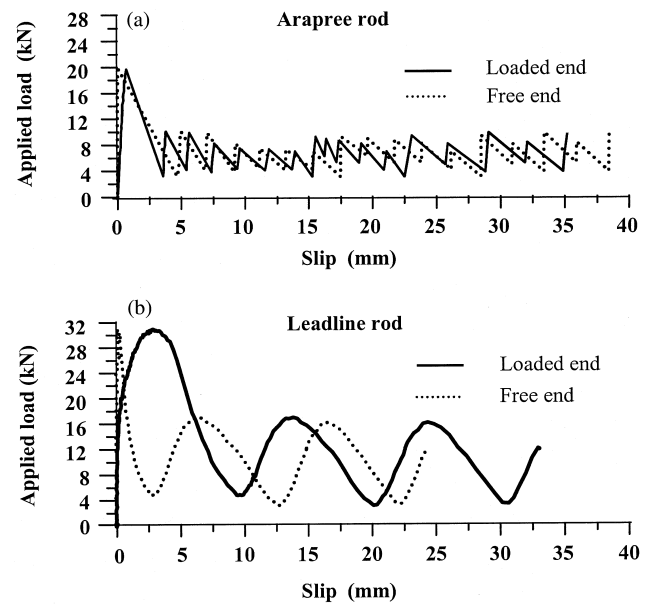


Fig. 9. Pullout behaviour of Arapree and Leadline rods.

also on the mechanical interlocking between the rod and grout.

3.4. Effect of rod type

The effect of different FRP rod types on the pullout behaviours is shown in Fig. 10. Different types of FRP rods have their own pullout behaviours with different pullout capacities and the corresponding slips. FRP rods with a larger deformation profile, such as Leadline, give higher maximum bond stress and initial stiffness in comparison with the other rods. The results shown in Tables 5 and 6 indicate that deformed AFRP Technora rods and CFRP CFCC and Leadline rods exhibit a higher maximum bond stress than epoxy-coated (a layer of sand and resin) AFRP Arapree rods. This can be explained by the fact that the deformed FRP rods develop maximum bond stress essentially by mechanical interlock mobilising the full column of cement grout in shear (bearing of the rod indentation against the grout). The deformed configuration of the

Table 5
Effect of bonded length on the maximum bond stress of FRP rods grouted in steel tubes with CG4

Tendon type	L = 100 mm		L = 200 mm		L = 350 mm	
	$T_{\max}$ (kN)	$\tau_{\max}$ (MPa)	$T_{\max}$ (kN)	$\tau_{\max}$ (MPa)	$T_{\max}$ (kN)	$\tau_{\max}$ (MPa)
Arapree-10	27.6 ± 4.0	8.8 ± 1.3	38.6 ± 4.6	6.1 ± 0.7	48.1 ± 4.8	4.4 ± 0.4
Arapree-8	20.6 ± 1.4	8.7 ± 0.6	26.9 ± 1.8	5.7 ± 0.4	37.1 ± 2.6	4.5 ± 0.3
Technora	33.3 ± 3.1	13.2 ± 1.2	55.6 ± 1.4	11.1 ± 0.3	67.9	7.7
				(2)	(1)	(1)
Leadline	35.8 ± 4.8	14.4 ± 1.9	67.6	13.6	T.F.	T.F.
			(1)	(1)		

Table 6

Effect of grout type on the maximum bond stress of FRP rods grouted in steel tubes^a

Tendon type	CG1		CG2		CG3		CG4	
	$T_{\max}$ (kN)	$\tau_{\max}$ (MPa)	$T_{\max}$ (kN)	$\tau_{\max}$ (MPa)	$T_{\max}$ (kN)	$\tau_{\max}$ (MPa)	$T_{\max}$ (kN)	$\tau_{\max}$ (MPa)
Arapree-8	19.4 ± 0.5	8.2 ± 0.2	18.6 ± 1.5	7.9 ± 0.6	19.9 ± 1.2	8.4 ± 0.5	20.6 ± 1.4	8.7 ± 0.6
Technora	30.9 ± 1.3	12.3 ± 0.5	20.0 ± 4.7	7.9 ± 1.9	31.1 ± 1.1	12.3 ± 0.4	33.3 ± 3.1	13.2 ± 1.2
Leadline	32.6 ± 5.2	13.1 ± 2.1	26.7 ± 1.8	10.6 ± 0.7	30.8 ± 5.8	12.4 ± 2.3	35.8 ± 4.8	14.4 ± 1.9
CFCC	31.1 ± 0.8	13.2 ± 0.3	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

^aNotes: The bonded length $L = 100$ mm; and n.a. = non-available.

ribs/indentations is an important parameter affecting both rib/indentation shear and grout shear. On the other hand, the coated rods develop their maximum bond stress mainly by the adhesion and frictional shear resistance at the rod–grout interface. The composition of the resin rich surface layer and its smoothness are the critical parameters [16,17]. This point was confirmed by visualising the failure modes at the rod–grout interface when rods were pulled out of the grout column [18]. Therefore, Arapree rods behave differently in terms of maximum bond stress due to their sand-coated surface in comparison with other deformed rods. Grouted in steel tubes with 100 mm bonded length, the average maximum bond stress of Arapree sand-coated rods is 7.9–8.8 MPa, 61–75% that of Leadline indented rods.

3.5. Effect of anchorage bonded length

Tests with varied anchorage bonded lengths were conducted to investigate the effect on the pullout capacity and bond behaviour. Fig. 11 shows the typical pullout load–slip curves on Arapree and Leadline rods. It indicates that the shape of the load–slip curves is similar, but both the pullout capacity and initial stiffness (slope of load–slip curve) increase with the increase in bonded length. An increase in bonded length

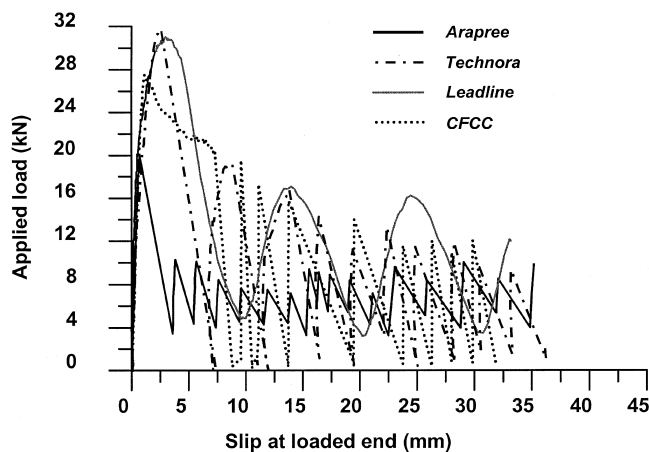


Fig. 10. Effect of the FRP rod type on the pullout behaviour.

results in decreasing the maximum bond stress, even though increasing the maximum pullout capacity, for all the tested FRP rods as shown in Table 5 and Fig. 12. This indicates that the bond stress distribution is not uniform along the bonded length for FRP rods. The longer the bonded length, the less uniformity the stress distribution. This agrees with the monitoring test results by Khin et al. [19].

3.6. Effect of cement grout

Fig. 13 shows typical effects of grout type on the pullout capacity and bond behaviour of FRP Leadline

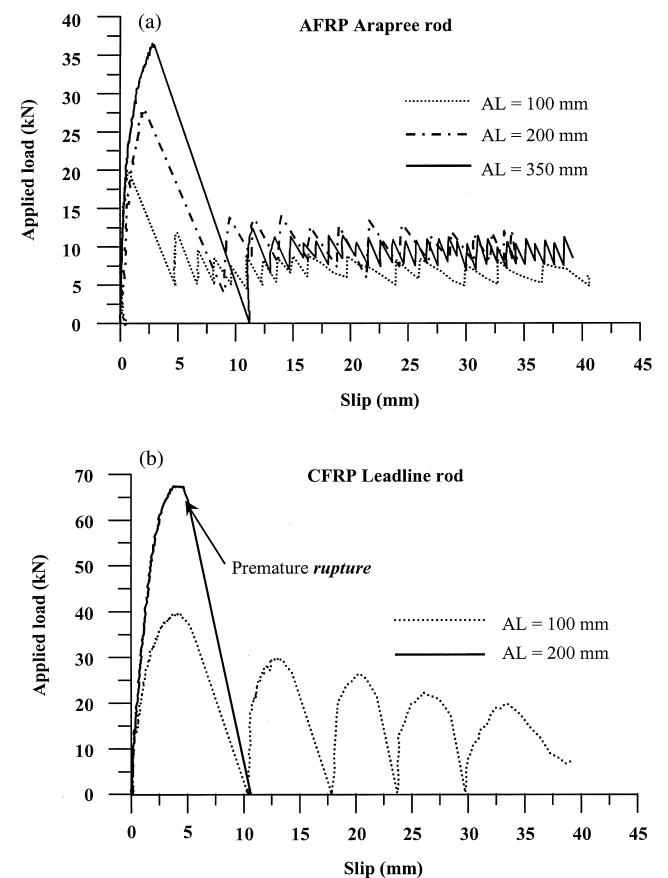


Fig. 11. Effect of bonded length on the pullout behaviour of (a) Arapree rod; and (b) Leadline rod.

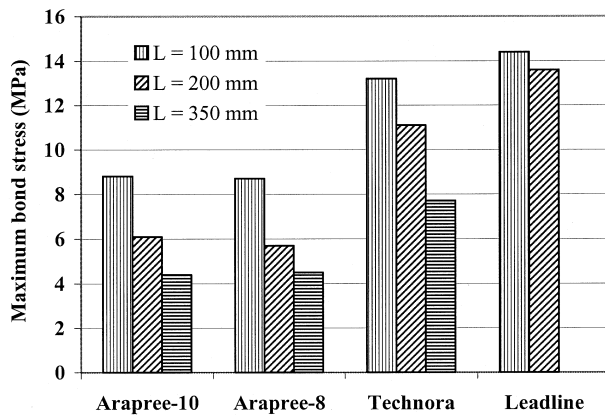


Fig. 12. Maximum bond stress of different FRP rods grouted with different bonded lengths.

rod. Anchor models with different cement grouts have different pullout capacities and bond behaviours. The properties of cement grouts influence the bond strength of all types of rods as shown in Fig. 14. Cement grouts CG3 and CG4 appear to give the best performance in terms of maximum bond stress, while cement CG2 presents the worst.

3.7. Effect of cement type

In comparison with grout CG1 made from Type 10 cement, grout CG2 made from Type 30 cement has higher compressive and tensile strength, and higher modulus of elasticity, but presents lower maximum

bond stress, as shown in Tables 2 and 6. In fact, maximum bond stress depends greatly on the condition and quality of the mated surfaces of the rod and grout. Type 30 cement is finer and has a higher C3A and C4AF content. This could have increased the autogenous shrinkage over that found with Type 10 cement [20]. The reduction in shear resistance of the rods embedded in grout CG2 is probably due to the autogenous shrinkage of this grout made from Type 30 cement. The shrinkage may be attributed to the reduction of mean size of capillary pores, the mean size in turn being affected by surface tension.

3.8. Effect of sand

It can be seen from Table 6 that the introduction of a proportional sand increases the maximum bond stress for all the anchors tested. This result agrees with the findings of previous studies of pullout tests on grouted FRP rods [16,21]. It can be said that the introduction of sand to the grout reduces its porosity and its shrinkage. On the other hand, the incompressibility of sand gives rise to dilatancy of cement grout at failure, leading to an improvement of frictional bond resistance. The pull-out force to the rod is balanced by the frictional resistance of the injected grout, so the sand content becomes important in maximising the frictional coefficient of the mortar grout. Test results as given in Table 2 confirm that the introduction of sand in grout

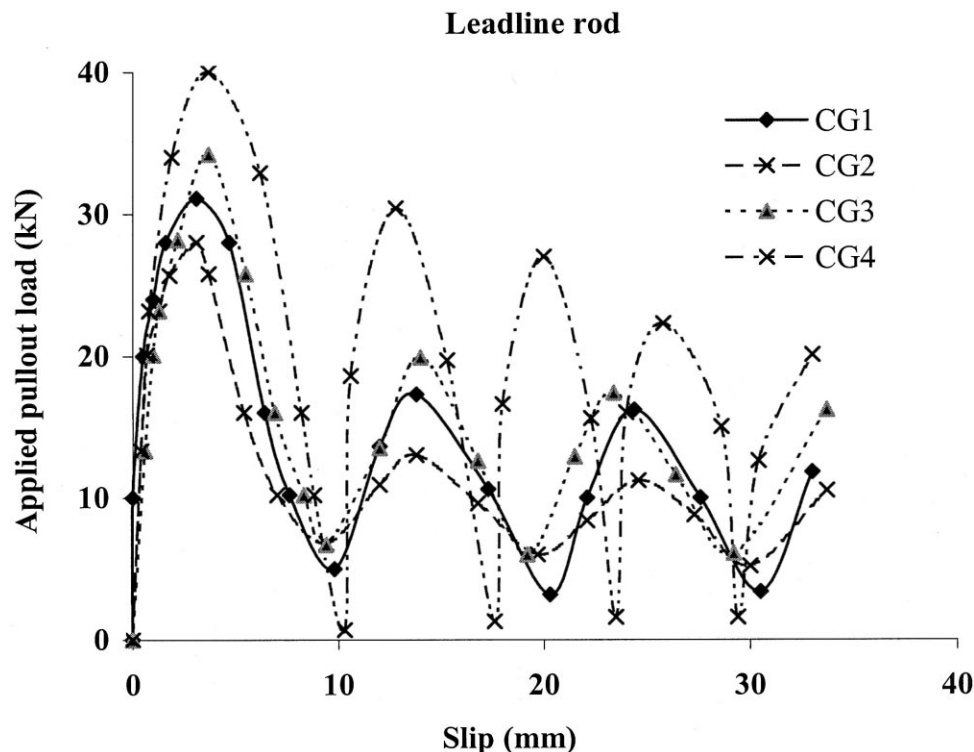


Fig. 13. Effect of grout type on the pullout behaviour of Leadline rod.

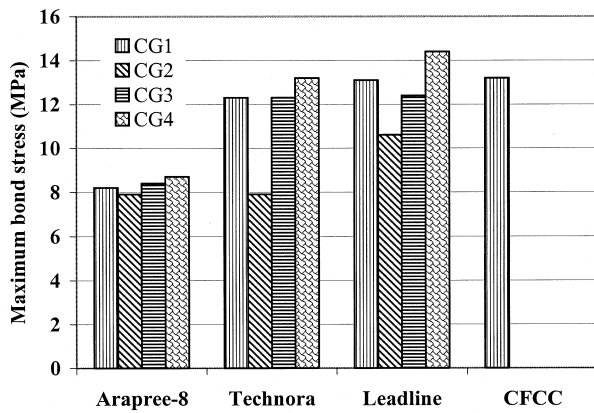


Fig. 14. Maximum bond stress of different FRP rods grouted with different cement grouts.

CG3 results in higher modulus of elasticity and friction angle in comparison with those of grout CG1. Earlier theoretical analysis on the parameters also indicates that an increase in the elastic modulus of the cement grout results in increasing the radial stiffness at the rod–grout interface and hence improves the maximum bond stress between them [22].

3.9. Effect of expansive agent

Table 2 indicates that the introduction of a swelling agent in grout CG4 decreases its compressive strength, modulus of elasticity and cohesion; but increases its friction angle and Poisson's ratio, in comparison with those of grout CG1. As seen in Table 6, the maximum bond stress is increased by using CG4 to bond the FRP rods. The comments could be made that the testing procedure for the grout with swelling agent may not be appropriate to the grout injected in a tube. Cast in the mold, the grout is free to expand in one direction. This is not completely true for grout injected in a tube or borehole, where great confinement is provided to the grout. Less expansion will therefore, take place and give a less porous grout and a lower reduction in strength. Moreover, the swelling agent creates swelling pressure during curing, which effectively improves the shear bond resistance at the rod–grout interface. Previ-

ous pullout test results have also indicated that using non-shrinking and expansive cement grouts can achieve a higher bond strength [19,23–26].

3.10. Effect of compressive strength

Test results shown in Tables 3 and 6 do not reveal the relationship between the compressive strength and the bond characteristics of the grout to the rod tested, since the grout constituent is different. However, previous bond tests on FRP rods embedded in concrete have shown a very low dependence of maximum bond stress on concrete strength in compression for smooth, twisted and sanded rods as well as the deformed rods obtained by gluing a fiber spiral on the outer surface [17,21,27,28]. The tests developed on deformed or indented rods show different bond mechanisms depending on the shape of the outer surface and the compressive strength of concrete. This last parameter influences the bond only when the ribs or indentations have a sufficient bearing strength [17,29]. When the rods have the right characteristics (i.e. high shear strength, high stiffness, optimum deformation geometry) to provide enough lateral confinement through ribs bearing, there is a dependence of maximum bond stress on compressive strength of concrete [28]; however, this influence is lower than the case of steel rebars since the deformations of FRP rods are shallower. The bond of deformed FRP rods with concrete is more dependent upon the friction and adhesion between the rod's surface and the concrete than on the mechanical interlocking between the rod deformations and the surrounding concrete.

3.11. Effect of host media

Fig. 15 shows the typical pullout behaviour of Leadline rods grouted in different types of anchorage tubes. Table 3 shows the modulus of elasticity of the equivalent rock medium for the three types of anchorage tubes. It can be concluded that the elastic modulus of host media (equivalent rock media) greatly influences the pullout behaviour of the grouted anchors. An in-

Table 7
Effect of anchorage tube on the maximum bond stress of FRP rods grouted with CG1^a

Rod type	Anchorage tube made of					
	Steel		Aluminium		PVC	
	(kN)	(MPa)	(kN)	(MPa)	(kN)	(MPa)
Arapree-8	19.4 ± 0.5	8.2 ± 0.2	16.7 ± 0.7	7.2 ± 0.3	13.5 ± 0.3	5.7 ± 0.1
Technora	30.9 ± 1.3	12.3 ± 0.5	27.2 ± 0.6	10.8 ± 0.3	13.1 ± 1.0	5.2 ± 0.4
Leadline	32.6 ± 5.2	13.1 ± 2.1	29.7 ± 1.2	12.0 ± 0.5	18.0 ± 2.1	7.3 ± 0.9
CFCC	31.1 ± 0.8	13.2 ± 0.3	27.8 ± 1.2	11.8 ± 0.5	24.0 ± 2.1	10.2 ± 0.9

^aNote: The bonded length $L = 100$ mm.

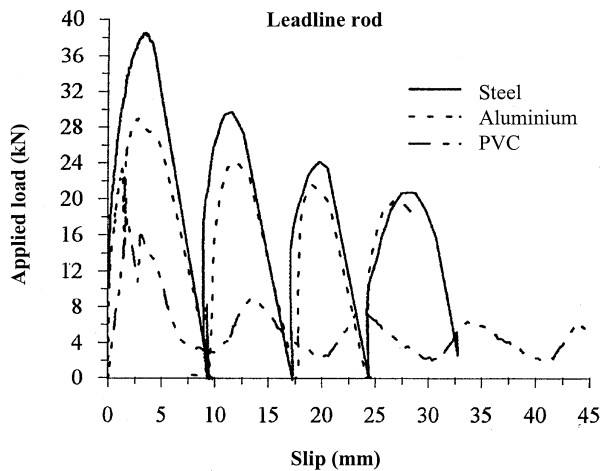


Fig. 15. Effect of anchorage tube type on the pullout behaviour of the Leadline rod.

crease in the modulus of elasticity of the medium results in increasing the pullout capacity, maximum bond stress and initial stiffness of the tested FRP anchors, as shown in Table 7 and Figs. 16 and 17. In fact, the media with higher elastic modulus increase confinement or radial stress on the grout. This prevents the propagation of cracks in the grout and as a result improves the pullout capacity of the anchors. Earlier theoretical research results have also shown that an increase in the dimension of the anchorage tube (a decrease in the stiffness of media) results in decreasing the radial stiffness; while an increase in the thickness of the tube (an increase in the stiffness of media) results in increasing the radial stiffness at the rod–grout interface. The effect is not significant when the stiffness of the host media is greater than a certain value [21].

4. Conclusions

In this investigation, experimental results on the

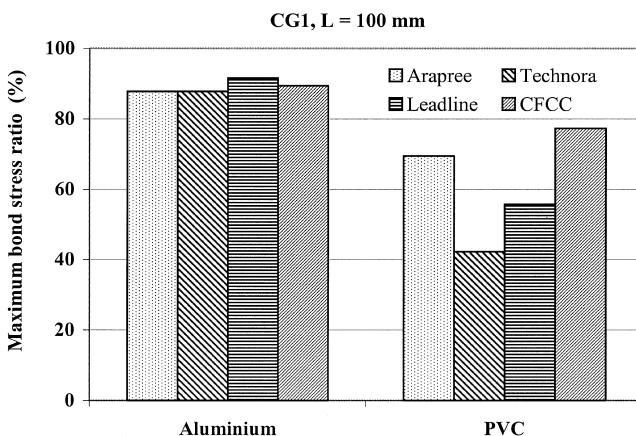


Fig. 16. Effect of anchorage tubes (Aluminium and PVC) in comparison with steel tubes on maximum bond stress.

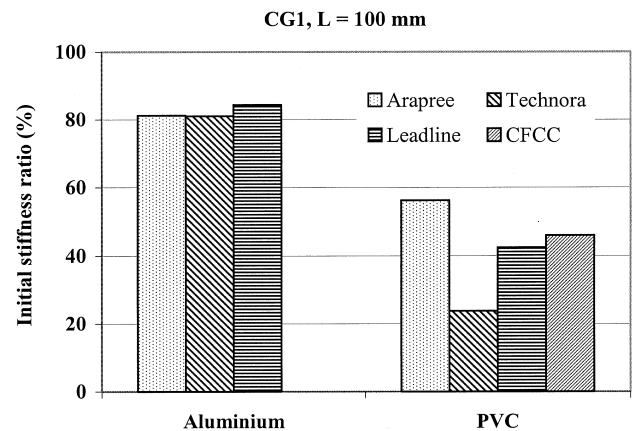


Fig. 17. Effect of anchorage tubes (aluminium and PVC) in comparison with steel tubes on initial stiffness.

evaluation of tensile characteristics, pullout capacity and bond properties of cement grouted FRP anchors are reported. These results provide experimental information for the design of bond-type anchorages for FRP rods, and also assist practising engineers in the design of FRP anchors for a wide range of applications in civil and mining engineering. Based on this study, the following conclusions can be drawn:

1. FRP rods are evaluated in tension to failure, except Technora rod which failed due to the fact that the anchorage system was only capable of attaining approximately 40% of the reported ultimate load of the rod. The manufacturers' values of strength, modulus of elasticity and ultimate strain appear to be in good agreement with experimentally determined results.
2. The pullout load–slip curves of grouted FRP anchors present ascending (pre-peak) and descending (post-peak) phases. In the pre-peak phase, the pullout load increases significantly with the increase in slip up to the peak load (maximum pullout capacity of the anchor). In the post-peak phase, grouted FRP anchors exhibit a sharp decrease in pullout capacity for a given increase in slip and thereafter present a lower and oscillating residual pullout capacity. The detachment of the fiber spiral or sand grains and the shearing off of the surface deformation typically cause the bond collapse of the FRP rod to grout. The bond is more dependent upon the adhesion and friction between the rod's surface and the grout than on the mechanical interlocking between them.
3. The bond behaviour and pullout capacity of grouted FRP anchors are influenced by many factors, mainly by the rod's surface deformation, properties of cement grouts, bonded length and the elastic modulus of host media (anchorage tubes). The indentation or surface deformation produced on the FRP

rods plays a major role in the maximum bond stress as well as the load–slip behaviour. FRP rods with a larger deformation profile, such as Leadline, give a higher maximum bond stress and stiffness in comparison with Arapree sand-coated rods. The average maximum bond stress of Leadline indented rods grouted in steel tubes with 100 mm bonded length is 10.6–14.4 MPa, 1.3–1.6 times that of Arapree sand-coated rods under the same conditions.

4. Bond collapse first occurs at the loaded end and then propagates toward the full bonded length. Bond stress distribution is not uniform along the bonded length. Maximum bond stress decreases with the increase in bonded length. The introduction of sand and some additives such as swelling agents to cement grouts appears to improve the maximum bond stress. The pullout capacity and stiffness of grouted FRP anchors also increase with the increase of the elastic modulus of host media.
5. The cement-grouted anchorages can provide an adequate and simple mechanism for anchoring the FRP rods discussed herein. The 350 mm steel tube anchorage with cement grout achieves the guaranteed tensile strength of the 8 mm diameter CFRP Leadline rods. The ultimate pullout capacity of grouted FRP anchors is generally controlled by the anchorage system rather than the rod itself, suggesting that anchor efficiency can be improved. It is proposed to use a bond anchorage system, consisting of a steel tube and a non-shrinking sand–cement grout, to anchor FRP rods. The minimum 28-day cylinder strength of the grout should be 50 MPa.

5. Notation

The following symbols are used in this paper:

A	=	tendon cross-section area;
d	=	tendon diameter;
E	=	modulus of elasticity;
f'_c	=	compressive strength of grout.
f_t	=	tensile strength of grout
k_i	=	initial stiffness;
l	=	tendon bonded length;
S	=	distance between the top of the bonded length and the measuring point;
T	=	applied tensioning load;
δ	=	bar slip along the bonded length;
δ_{mea}	=	measured total displacement;
δ_{bar}	=	bar elastic elongation; and
τ	=	bond stress
ν	=	Poisson's ratio.

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